PA2 Task 1B: Stride Pattern Prefetcher

CS683

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1 Experimental setup

Task 1B asks for IPC, speedup, L1D load MPKI, prefetch accuracy, and L1D coverage for three traces, together with plots. It also asks for an exact storage breakdown and the rationale for the accuracy thresholds and their prefetch degrees. The simulations reported here use a 25-million-instruction warmup and a 25-million-instruction region of interest per trace, on a single-core ChampSim configuration with a hashed-perceptron branch predictor and LRU replacement. The baseline uses no L1D prefetcher; the other runs use IP-stride and the implemented SPP.

2 Measured results

Speedup is calculated as the prefetcher IPC divided by the no-prefetch baseline IPC. MPKI and accuracy are taken from ChampSim's L1D load and prefetch statistics. Coverage uses the assignment's definition:

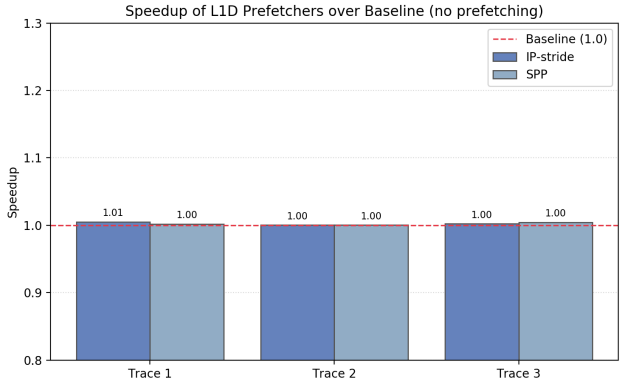
$$\text{coverage} = \frac{\text{baseline L1D load misses} - \text{prefetcher L1D load misses}}{\text{baseline L1D load misses}} \times 100\%.$$

For the baseline, accuracy and coverage are not applicable.

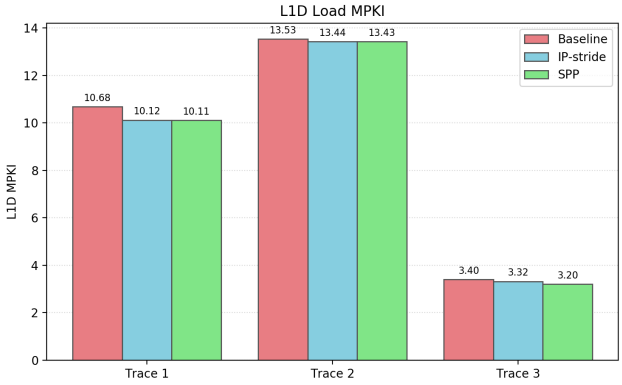
Table 1: Measured IPC and prefetcher metrics for each trace.

Trace	Configuration	IPC	Speedup	L1D load MPKI	Accuracy (%)	Coverage (%)
Trace 1	Baseline	0.232204	1.000000	10.6787	–	–
	IP-stride	0.233398	1.005142	10.1188	52.7914	5.2433
	SPP	0.232646	1.001903	10.1085	37.2389	5.3400
Trace 2	Baseline	0.251130	1.000000	13.5338	–	–
	IP-stride	0.251186	1.000223	13.4356	38.1276	0.7253
	SPP	0.251183	1.000211	13.4314	21.0828	0.7563
Trace 3	Baseline	0.382061	1.000000	3.4032	–	–
	IP-stride	0.382962	1.002358	3.3177	16.0936	2.5141
	SPP	0.383744	1.004405	3.1980	23.7845	6.0319

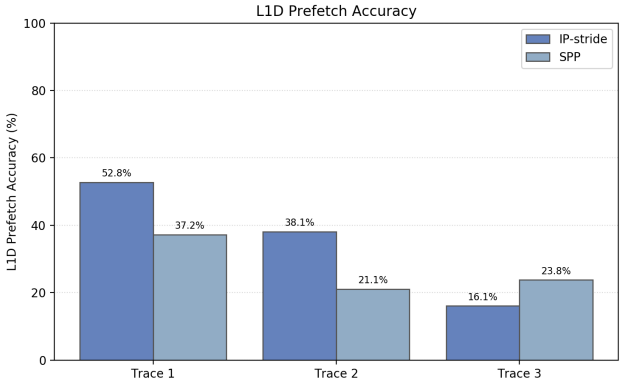
Both prefetchers give small IPC improvements in these runs: IP-stride ranges from 1.000223x to 1.005142x, and SPP ranges from 1.000211x to 1.004405x. SPP produces the lowest MPKI on all three traces, although its miss reduction is modest. In particular, its coverage is below 1% on Trace 2 and about 5–6% on Traces 1 and 3. These are the measured results under the stated coverage formula; the higher bar heights in the assignment's example figures are illustrative, not values to substitute for measurements.



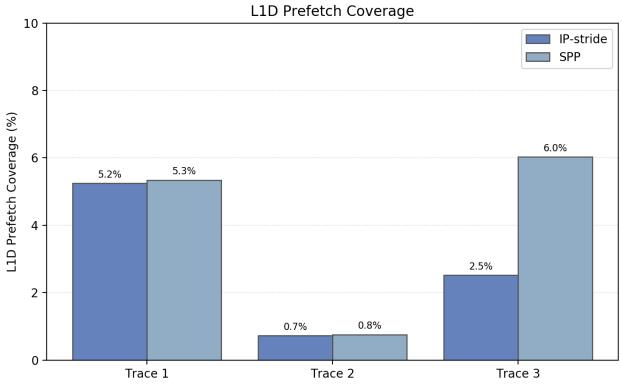
(a) IPC speedup over the no-prefetch baseline.



(b) L1D load MPKI.



(c) Prefetch accuracy reported by ChampSim.



(d) L1D load-miss reduction relative to baseline.

Figure 1: Task 1B metrics across the three traces. The coverage plot uses a 0–10% vertical range so that the measured single-digit miss reductions remain readable without changing or inflating the values.

3 SPP design and storage overhead

The SPP is implemented at the L1D. It tracks accesses by a 12-bit partial IP tag and keeps the page tag, line offset, last stride, uniform-stride confidence, and a 12-bit signature. The signature is updated with the low four bits of each observed stride and indexes a 128-entry pattern table after folding. A matching pattern-table entry predicts the next stride. The prefetcher uses the uniform predictor after three consecutive equal strides; otherwise, it uses a sufficiently confident pattern-table prediction. Prefetches are restricted to the current 4 KB page. It chooses L1 or L2 as the fill target based on MSHR occupancy.

The following field widths are the storage widths of the declared C++ types (`bool`=8, `uint8_t`/`int8_t`=8, and `uint16_t`=16 bits). They therefore account for the implementation as written, not for hypothetical bit-packing.

Table 2: SPP table entries and storage, including compiler padding.

Table	Fields per entry	Entries	Declared field bits	Allocated bits
IP tracker	valid (8), IP (16), page tag (16), line offset (8), last stride (8), uniform confidence (8), signature (16), LRU (8)	32	88	112
Pattern table	next stride (8), confidence (8)	128	16	16
Prefetch filter	valid (8), tag (16)	32	24	32
Tables total			5,632	6,656

On the GCC 7.5 x86-64 build used here, `sizeof(IP_TRACKER)` is 14 bytes, `sizeof(PT_ENTRY)` is 2 bytes, and `sizeof(PF_ENTRY)` is 4 bytes. Thus the tables occupy 832 bytes (6,656 bits), including 128 bytes of

structure padding. The four 64-bit statistics counters and the three 32-bit epoch/degree counters occupy another 44 bytes (352 bits). Counting both tables and these implementation counters, the SPP state is 876 bytes (7,008 bits), excluding code, cache-line fill storage, and simulator-owned metadata.

4 Accuracy throttling

The prefetcher starts with degree 2 and accumulates accepted prefetches and subsequent useful demand accesses in a direct-mapped recent-prefetch filter. After at least 128 accepted prefetches, it computes the integer percentage of useful prefetches in the epoch and selects a degree for subsequent requests:

Epoch accuracy	Degree
$< 25\%$	1
$25\%– < 50\%$	2
$50\%– < 75\%$	4
$\geq 75\%$	6

The 128-request epoch smooths individual outcomes while allowing the degree to respond during a long trace. The thresholds favor conservatism when predictions are often unused, retain modest lookahead at intermediate accuracy, and permit greater lookahead when predictions are consistently useful. The degree is capped at 6 to limit bandwidth and cache-pollution risk. The prefetcher starts at degree 2 so it can collect useful training and accuracy observations before the first adjustment.

These are heuristic design thresholds, not the result of a threshold sensitivity sweep. The assignment requests the reasoning behind the chosen thresholds and degrees; the runs in this report evaluate the resulting configuration but do not claim to have empirically optimized those thresholds.